SECTION 1: DETAILED MULTI-TIER SYSTEM ARCHITECTURE

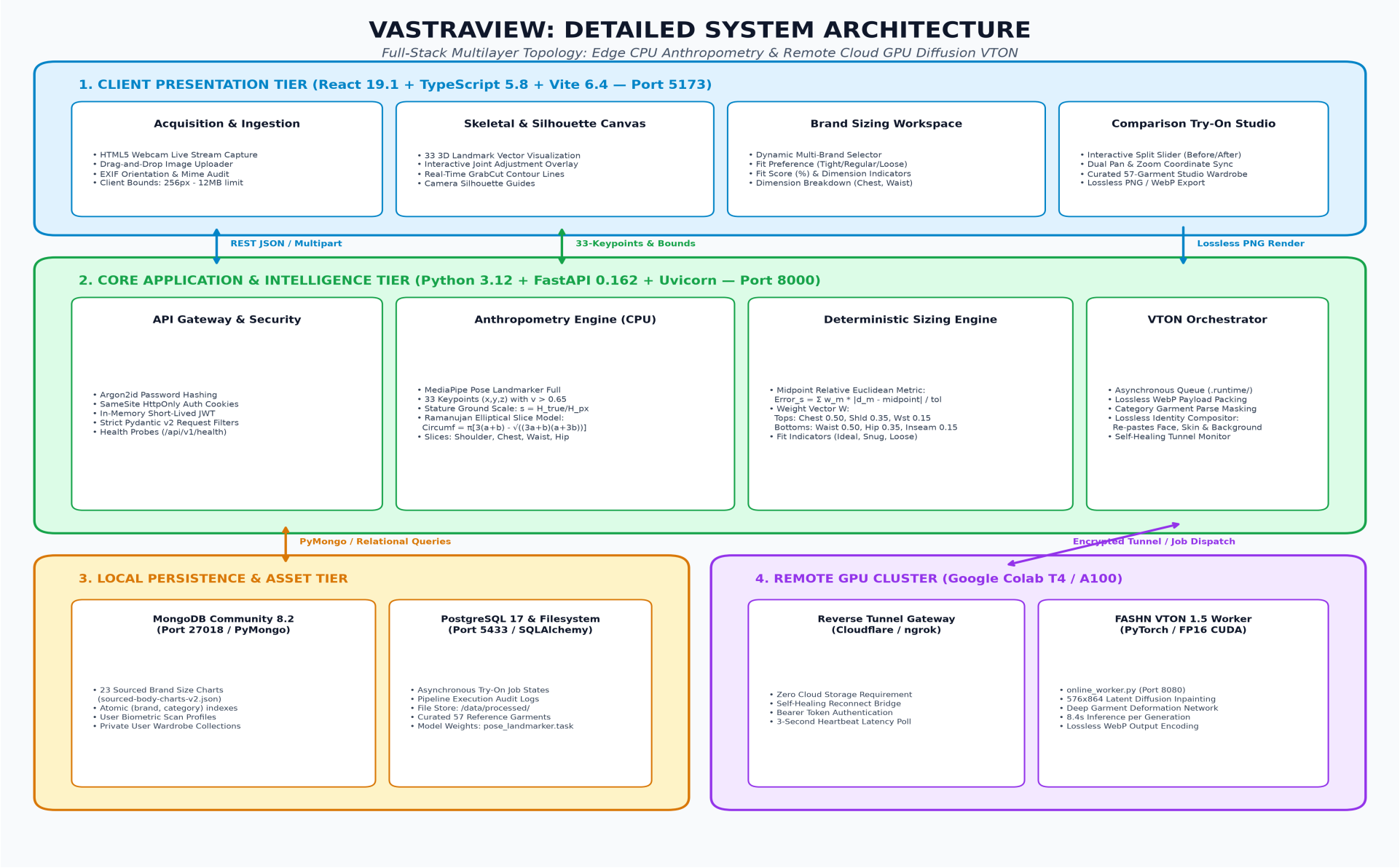


Figure 1.1: Multi-Tier Architectural Topology. Depicts the physical and logical boundaries separating client-side presentation (React 19), deterministic edge CPU anthropometry (FastAPI / MediaPipe), local databases (PostgreSQL 17 / MongoDB 8.2), and the remote Google Colab GPU diffusion worker.

SECTION 2: FUNCTIONAL PIPELINE & DATA-FLOW BLOCK DIAGRAM

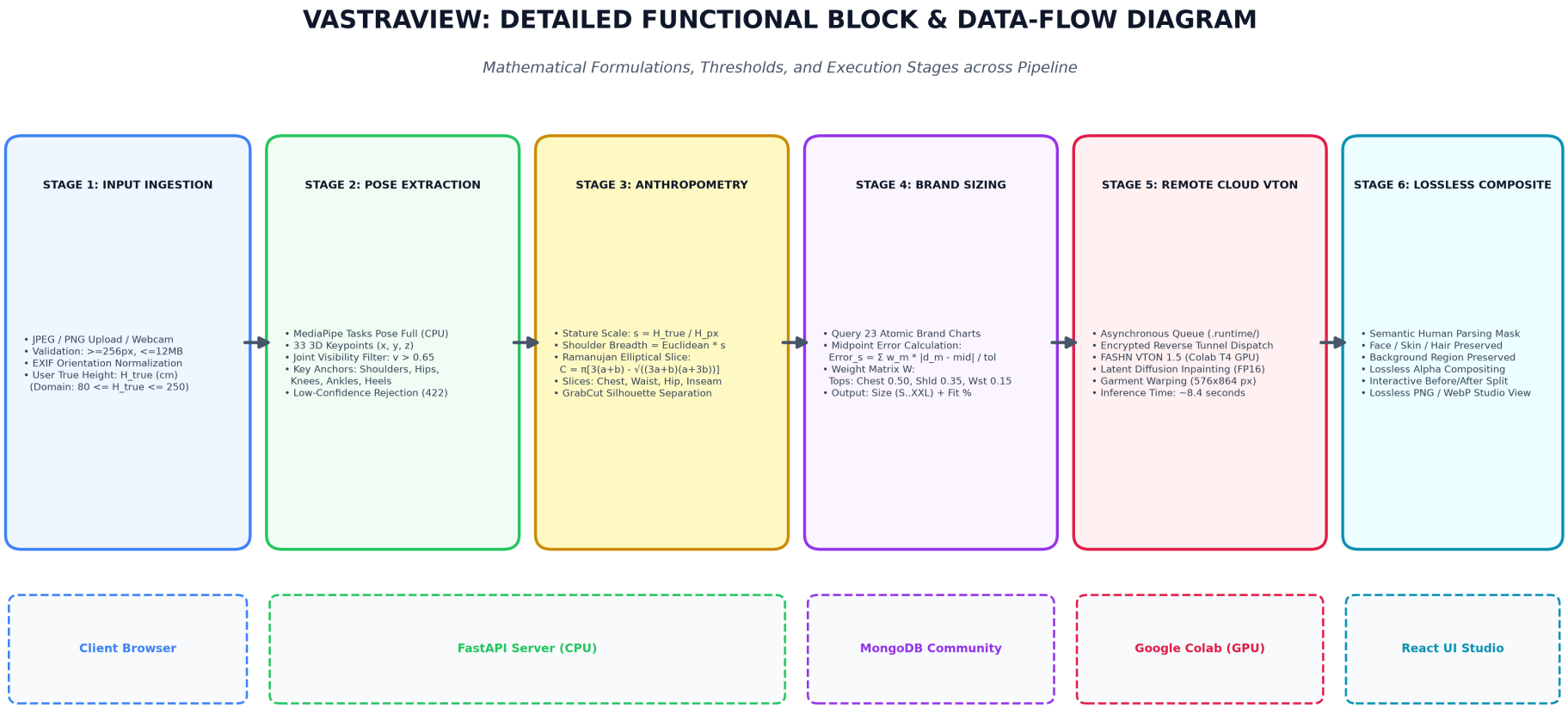


Figure 1.2: End-to-End Functional Data Flow. Details the 6 operational pipeline stages: image validation, 33-point CPU pose tracking, height-calibrated Ramanujan circumference computation, weighted brand chart matching, remote cloud GPU diffusion inpainting (FASHN VTON 1.5), and lossless identity composite output.

SECTION 3: ARCHITECTURAL EQUATIONS, DATAFLOWS & SUBSYSTEM CONTRACTS

Subsystem	Implementation Details	Mathematical Models & Thresholds	Hardware Target
1. Client Tier	React 19.1, TypeScript 5.8, Vite 6.4 HTML5 Canvas & WebRTC Camera Stream Split-Screen Comparison Slider	• Input resolution: 256px min, 12MB max • Viewport sync: dual pan/zoom coordinates • Export format: 24-bit sRGB PNG / WebP	Client Browser (Desktop / Mobile)
2. Anthropometry Engine	FastAPI, MediaPipe Tasks Pose Full GrabCut Foreground Segmentation Deterministic coordinate transformation	• Visibility acceptance threshold: v > 0.65 • Stature scaling factor: s = H_true / |y_ground - y_crown| • Ramanujan Ellipse: C ≈ π[3(a+b) - √((3a+b)(a+3b))]	Local Server CPU (No GPU required)
3. Sizing Engine	Atomic MongoDB Repository 23 Sourced Brand Chart fixtures (Nike, Adidas, Puma, Levi's, H&M, Manyavar)	• Weighted Error: Error_s = Σ w_m * |d_m - midpoint| / tol • Active Weights: Tops {Chest:0.50, Shld:0.35, Wst:0.15} • Bottoms {Waist:0.50, Hip:0.35, Inseam:0.15}	Local Server CPU (MongoDB 27018)
4. Remote Cloud GPU	FASHN VTON 1.5 Diffusion Model PyTorch, CUDA 12.x, FP16 Inpainting Self-healing reverse encrypted tunnel	• Resolution: 576 x 864 pixels • Classifier-free guidance: 1.5, Latency: ~8.4s • Lossless composite: FaceMask & Background preserved	Remote Google Colab (NVIDIA T4 / A100 GPU)